Deon Levon Dmello

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Experience

AI Engineer I April 2026 – Present

Bayer *Bengaluru, KA*

- Built GPT-5 (Azure OpenAI) agentic workflows with RAG for SAP S/4HANA procurement approvals, cutting approval time from days to minutes.
- Built backend services and REST APIs for agent orchestration and SAP data integration, deployed to AWS ECS Fargate via CI/CD workflows and used by 200–300 users.
- Added deterministic guardrails and validation so agent outputs stay correct and auditable across approvals of 5,000+ line-item changes.
- Automated header and line-item updates for the entire procurement team, cutting turnaround from a full day to minutes.
- Led API testing and QA for production agentic workflows.

AI Engineer June 2024 – March 2026

Tardid Technologies *Bengaluru, KA*

- **LiDAR–Camera Calibration:** Developed an automated tool for extrinsic calibration between LiDAR and camera sensors, improving spatial alignment accuracy and reducing manual calibration effort.
- **Real-time & Zero-Shot Action Recognition:** Developed and deployed action recognition systems using X3D, Swin3D, and X-CLIP for zero-shot inference, achieving 30 FPS on live RTSP streams.
- **Vision-Language Models (VLMs):** Integrated CLIP- and X-CLIP-based pipelines for cross-modal image–text retrieval, zero-shot classification, and scene understanding; fine-tuned CNN classifiers for a 10% F1-score gain on real-world datasets.
- **Large Language Models (LLMs):** Deployed and optimized instruction-tuned LLMs for vision-assisted analytics and system control using llama.cpp and SGLang, enabling low-latency inference on edge GPUs.
- **Backend Development:** Designed and maintained FastAPI-based backend APIs for ML inference, model orchestration, and data pipelines.

SRE / Platform Engineer Intern March 2024 – June 2024

Practo Technologies *Bengaluru, KA*

- Managed and resolved support tickets using JIRA, connecting to servers over SSH to troubleshoot and fix issues, ensuring system reliability and uptime.
- Implemented bug patches, enhanced platform stability, and developed a thorough understanding of in-house tools for sending SMS and email notifications.
- Managed backend microservices using Docker, ensuring efficient deployment, scalability, and reliability of platform services.
- Performed scheduled database backups and managed production Postgres dumps to support data recovery and environment migrations.
- Developed a data pipeline to collect and analyze logs from AWS CloudWatch, enabling proactive identification of errors and deployment issues.

Education

Indian Institute of Technology Madras Chennai, TN

BS in Programming and Data Science *September 2021 – Present*

- Completed Diploma in Data Science & Diploma in Programming

Reva University Bengaluru, KA

B.Tech Computer Engineering (AI & ML) *December 2020 – July 2024*

Projects

Sync'em | *FastAPI, Vue.js, TailwindCSS, PostgreSQL, Celery, Redis, GenAI Agents, Jira* | GitHub September 2025

- Built a full-stack workforce and project management platform in an Agile Scrum environment with a cross-functional 8-member team, delivering role-based portals for Admins, HR, Project Managers, and Employees.
- Implemented Generative AI assistants (LLM + RAG) to automate HR queries, onboarding workflows, and requirement clarification, significantly reducing manual support effort.
- Integrated asynchronous job pipelines with Celery and Redis for automated email notifications, scheduled reports, and background processing.
- Documented APIs using OpenAPI (Swagger) and wrote Pytest test suites for critical services, improving system reliability.

QuizNexus | *Vue.js, Flask, SQLite, Bootstrap, JWT, Redis, LangChain, LangGraph* | GitHub July 2025

- Developed a full-stack quiz platform with a Flask REST API and modular Vue.js frontend for dynamic quiz creation, administration, and responsive UI design.
- Built an AI-powered student advisor using LangChain and LangGraph to analyze quiz history, detect weak areas, and generate personalized study recommendations.
- Implemented secure role-based authentication using JWT, Redis caching for session optimization, and MailHog/SMTP for automated email verification.

ImageColorization | *Python, PyTorch, CNN, UNet, GANs* | GitHub April 2024

- Developed a CNN-based model using PyTorch to automatically colorize grayscale images, applying UNet, GANs, convolutional layers, and loss optimization techniques.
- Packaged the model into a documented Python module for usability and portability.

Technical Skills

Languages: Python, C/C++, Bash, SQL, Go
Databases & Storage: PostgreSQL, Redis, Object Storage
Machine Learning & Deep Learning: scikit-learn, PyTorch, OpenCV, CLIP
GenAI & RAG Systems: Azure OpenAI (GPT-5), LangChain, LangGraph, CrewAI, HuggingFace, RAG Pipelines
Vector Stores & Embeddings: FAISS, Qdrant
MLOps & Model Serving: FastAPI, REST APIs, Model Orchestration, CI/CD, Edge Deployment, GPU Inference, llama.cpp, SGLang, Quantized LLMs
Cloud & Reliability: AWS (ECS Fargate, EC2, CloudWatch), PagerDuty, Jira, SSH, Linux Server Administration
Enterprise Systems: SAP S/4HANA
Developer Tools: Docker, Git/Gitea, VS Code, Jupyter, Colab, Neovim
Operating Systems: Linux (Ubuntu), Windows

Certifications

- Advance Certificate in Programming and Application Development — IIT Madras
- Foundational Level Certificate in Programming and Data Science — IIT Madras
- TensorFlow Developer Certificate